

```
In [13]: # print()
```

```
In [15]: num1 = 20
num2 = 30
add = num1+num2
print(add)
```

50

```
In [18]: num1=2
num2=1000000000000000000
add = num1+num2
print('The addition of',num1,'and' ,num2, 'is=', add)
```

The addition of 2 and 1000000000000000000 is= 1000000000000000002

```
In [20]: #the adition of 20 and 30 is = 50
```

```
In [22]: num1 = 30
num2 = 30
num3 = 50
add = num1+num2+num3
print(add)
```

110

```
In [23]: name = 'python'
age = 20
city = 'hyd'
print('my name is',name, 'and i am',age,'years old form',city)
```

my name is python and i am 20 years old form hyd

```
In [26]: num1 = 20
num2 = 30
num3 = 5000
add = num1+num2
print(add)
```

50

```
In [31]: print('hello team',end = '')
print('good moorning')
```

hello teamgood moorning

```
In [32]: print('hello')
print('good moorning')
```

hello
good moorning

```
In [33]: print('hello team', end = '###')
print('good moorning')
```

hello team###good moorning

```
In [34]: #numpy crash course
```

```
In [48]: import numpy as np
```

```
In [50]: np.__version__
```

```
Out[50]: '2.1.3'
```

```
In [45]: my_list = [0,1,2,3,4,5]
         my_list
```

```
Out[45]: [0, 1, 2, 3, 4, 5]
```

```
In [49]: arr = np.array(my_list)
         arr
```

```
Out[49]: array([0, 1, 2, 3, 4, 5])
```

```
In [52]: print(type(my_list))
         print(type(arr))

<class 'list'>
<class 'numpy.ndarray'>
```

```
In [55]: np.arange(10,50,5)
```

```
Out[55]: array([10, 15, 20, 25, 30, 35, 40, 45])
```

```
In [56]: np.arange(20,10)
```

```
Out[56]: array([], dtype=int64)
```

```
In [57]: np.arange(-20,10)
```

```
Out[57]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
                -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
                 6,  7,  8,  9])
```

```
In [58]: np.zeros(3,dtype = float)
```

```
Out[58]: array([0., 0., 0.])
```

```
In [59]: np.zeros(3,dtype=int)
```

```
Out[59]: array([0, 0, 0])
```

```
In [60]: np.zeros([2,2])
```

```
Out[60]: array([[0., 0.],
                [0., 0.]])
```

```
In [61]: np.zeros([10,10],dtype=int)
```

```
Out[61]: array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

```
In [62]: np.ones(4)
```

```
Out[62]: array([1., 1., 1., 1.])
```

```
In [64]: np.random.rand(3)
```

```
Out[64]: array([0.82021756, 0.60178769, 0.4057235 ])
```

```
In [65]: np.random.rand(3,4)
```

```
Out[65]: array([[0.53577611, 0.52742683, 0.37250906, 0.69240135],
               [0.75309807, 0.76786712, 0.05338421, 0.31741635],
               [0.5426426 , 0.35902309, 0.47646418, 0.70083872]])
```

```
In [70]: np.random.rand(3)
```

```
Out[70]: array([0.17377303, 0.25103918, 0.21253348])
```

```
In [71]: np.random.rand(3,4)
```

```
Out[71]: array([[0.24689986, 0.50220795, 0.57446559, 0.22838004],
               [0.54787864, 0.18974389, 0.06637402, 0.36012021],
               [0.63712294, 0.7534526 , 0.99780276, 0.47834842]])
```

```
In [72]: np.random.rand(3)
```

```
Out[72]: array([0.14079357, 0.35663494, 0.17344405])
```

```
In [75]: np.random.randint(1,10,4)
```

```
Out[75]: array([6, 9, 7, 7], dtype=int32)
```

```
In [76]: print(np.random.randint(1,10,4))
```

```
[8 6 1 7]
```

```
In [77]: m = np.random.randint(1,10,(4,6))
         print(m)
```

```
[[1 4 5 1 2 7]
 [2 4 7 3 2 8]
 [4 9 9 9 6 5]
 [7 8 1 3 9 3]]
```

```
In [78]: #indexing
```

```
In [79]: print(m)
```

```
[[1 4 5 1 2 7]
 [2 4 7 3 2 8]
 [4 9 9 9 6 5]
 [7 8 1 3 9 3]]
```

```
In [80]: m[0]
```

```
Out[80]: array([1, 4, 5, 1, 2, 7], dtype=int32)
```

```
In [82]: m[0] = [4,5,6,7,8,9]
```

```
In [83]: m
```

```
Out[83]: array([[4, 5, 6, 7, 8, 9],
                [2, 4, 7, 3, 2, 8],
                [4, 9, 9, 9, 6, 5],
                [7, 8, 1, 3, 9, 3]], dtype=int32)
```

```
In [84]: m[2]
```

```
Out[84]: array([4, 9, 9, 9, 6, 5], dtype=int32)
```

```
In [85]: m[-3]
```

```
Out[85]: array([2, 4, 7, 3, 2, 8], dtype=int32)
```

```
In [86]: m
```

```
Out[86]: array([[4, 5, 6, 7, 8, 9],
                [2, 4, 7, 3, 2, 8],
                [4, 9, 9, 9, 6, 5],
                [7, 8, 1, 3, 9, 3]], dtype=int32)
```

```
In [87]: print(m[0,4])
```

8

```
In [88]: m
```

```
Out[88]: array([[4, 5, 6, 7, 8, 9],
                [2, 4, 7, 3, 2, 8],
                [4, 9, 9, 9, 6, 5],
                [7, 8, 1, 3, 9, 3]], dtype=int32)
```

```
In [89]: print(m[-2,-3])
```

9

```
In [90]: np.random.randint(10,40,(10,10))
```

```
Out[90]: array([[31, 32, 28, 18, 31, 37, 13, 32, 31, 38],
                [18, 23, 29, 20, 36, 30, 27, 34, 26, 15],
                [10, 30, 32, 13, 30, 30, 26, 33, 36, 14],
                [15, 16, 35, 11, 11, 33, 12, 32, 18, 16],
                [38, 31, 24, 20, 18, 33, 32, 19, 25, 24],
                [22, 17, 10, 31, 26, 16, 10, 23, 19, 33],
                [34, 18, 22, 35, 35, 11, 24, 11, 32, 31],
                [29, 16, 11, 33, 28, 39, 34, 35, 19, 28],
                [27, 21, 13, 12, 16, 27, 16, 16, 23, 11],
                [23, 13, 37, 12, 15, 39, 24, 21, 26, 15]], dtype=int32)
```

```
In [91]: np.random.randint(1,100,(12,12))
```

```
Out[91]: array([[39, 46, 22, 76, 33, 54, 76, 83, 41, 62, 33, 92],
 [78, 36, 26, 59, 83, 37, 72, 83, 95, 75, 18, 73],
 [87, 18,  1, 75, 44, 62, 31, 77, 74,  5, 27, 41],
 [45, 49, 36, 63, 26, 58, 29, 39, 33, 35, 46, 38],
 [95, 62,  5, 79, 98, 48, 44, 99, 95, 42, 30,  3],
 [51, 23,  7, 43, 55, 35, 95, 23, 34,  3, 52, 23],
 [19, 25, 97, 40, 11, 65, 88, 79, 36, 86, 85, 82],
 [77, 67, 83, 14, 22, 88, 72, 47, 17, 48, 16, 89],
 [45, 31, 29, 83, 95, 71,  1, 14, 11, 64, 34, 16],
 [57, 27, 25, 20,  1, 93, 94, 62, 66, 39, 29, 83],
 [47, 20, 66, 61, 25, 82, 31, 37, 10, 81, 94,  3],
 [66, 60, 31,  9, 22, 33, 49, 59, 42,  8, 50, 53]], dtype=int32)
```

```
In [93]: np.arange(1,13).reshape(3,4)
```

```
Out[93]: array([[ 1,  2,  3,  4],
 [ 5,  6,  7,  8],
 [ 9, 10, 11, 12]])
```

```
In [94]: np.arange(1,13).reshape(12,1)
```

```
Out[94]: array([[ 1],
 [ 2],
 [ 3],
 [ 4],
 [ 5],
 [ 6],
 [ 7],
 [ 8],
 [ 9],
 [10],
 [11],
 [12]])
```

```
In [96]: b = np.random.randint(10,20,(5,4))
b
```

```
Out[96]: array([[14, 17, 13, 17],
 [16, 11, 14, 19],
 [10, 10, 11, 19],
 [15, 14, 19, 10],
 [18, 10, 19, 11]], dtype=int32)
```

```
In [97]: type(b)
```

```
Out[97]: numpy.ndarray
```

```
In [98]: b
```

```
Out[98]: array([[14, 17, 13, 17],
 [16, 11, 14, 19],
 [10, 10, 11, 19],
 [15, 14, 19, 10],
 [18, 10, 19, 11]], dtype=int32)
```

```
In [99]: b[:]
```

```
Out[99]: array([[14, 17, 13, 17],
               [16, 11, 14, 19],
               [10, 10, 11, 19],
               [15, 14, 19, 10],
               [18, 10, 19, 11]], dtype=int32)
```

```
In [100... b[1:3]
```

```
Out[100... array([[16, 11, 14, 19],
                  [10, 10, 11, 19]], dtype=int32)
```

```
In [101... b
```

```
Out[101... array([[14, 17, 13, 17],
                  [16, 11, 14, 19],
                  [10, 10, 11, 19],
                  [15, 14, 19, 10],
                  [18, 10, 19, 11]], dtype=int32)
```

```
In [102... b[1,2]
```

```
Out[102... np.int32(14)
```

```
In [103... b
```

```
Out[103... array([[14, 17, 13, 17],
                  [16, 11, 14, 19],
                  [10, 10, 11, 19],
                  [15, 14, 19, 10],
                  [18, 10, 19, 11]], dtype=int32)
```

```
In [104... b[1,3]
```

```
Out[104... np.int32(19)
```

```
In [105... b[1,-1]
```

```
Out[105... np.int32(19)
```

```
In [106... b
```

```
Out[106... array([[14, 17, 13, 17],
                  [16, 11, 14, 19],
                  [10, 10, 11, 19],
                  [15, 14, 19, 10],
                  [18, 10, 19, 11]], dtype=int32)
```

```
In [107... b[2:3]
```

```
Out[107... array([[10, 10, 11, 19]], dtype=int32)
```

```
In [108... b
```

```
Out[108... array([[14, 17, 13, 17],
                  [16, 11, 14, 19],
                  [10, 10, 11, 19],
                  [15, 14, 19, 10],
                  [18, 10, 19, 11]], dtype=int32)
```

```
In [109... b[0:-2]
```

```
Out[109... array([[14, 17, 13, 17],  
        [16, 11, 14, 19],  
        [10, 10, 11, 19]], dtype=int32)
```

```
In [110... b
```

```
Out[110... array([[14, 17, 13, 17],  
        [16, 11, 14, 19],  
        [10, 10, 11, 19],  
        [15, 14, 19, 10],  
        [18, 10, 19, 11]], dtype=int32)
```

```
In [111... b[0,2]
```

```
Out[111... np.int32(13)
```

```
In [112... b
```

```
Out[112... array([[14, 17, 13, 17],  
        [16, 11, 14, 19],  
        [10, 10, 11, 19],  
        [15, 14, 19, 10],  
        [18, 10, 19, 11]], dtype=int32)
```

```
In [113... b[-5,-3]
```

```
Out[113... np.int32(17)
```

```
In [114... b
```

```
Out[114... array([[14, 17, 13, 17],  
        [16, 11, 14, 19],  
        [10, 10, 11, 19],  
        [15, 14, 19, 10],  
        [18, 10, 19, 11]], dtype=int32)
```

```
In [115... b[-4,2]
```

```
Out[115... np.int32(14)
```

```
In [116... b
```

```
Out[116... array([[14, 17, 13, 17],  
        [16, 11, 14, 19],  
        [10, 10, 11, 19],  
        [15, 14, 19, 10],  
        [18, 10, 19, 11]], dtype=int32)
```

```
In [117... b[-4,-2]
```

```
Out[117... np.int32(14)
```

```
In [118... b
```

```
Out[118...] array([[14, 17, 13, 17],
          [16, 11, 14, 19],
          [10, 10, 11, 19],
          [15, 14, 19, 10],
          [18, 10, 19, 11]], dtype=int32)
```

```
In [119...] b[:]
```

```
Out[119...] array([[14, 17, 13, 17],
          [16, 11, 14, 19],
          [10, 10, 11, 19],
          [15, 14, 19, 10],
          [18, 10, 19, 11]], dtype=int32)
```

```
In [120...] #operations
```

```
In [121...] a = np.random.randint(10,20,10)
a
```

```
Out[121...] array([19, 11, 15, 14, 11, 18, 11, 12, 12, 12], dtype=int32)
```

```
In [122...] id(a)
```

```
Out[122...] 3076942893904
```

```
In [123...] arr
```

```
Out[123...] array([0, 1, 2, 3, 4, 5])
```

```
In [ ]:
```

```
In [127...] arr2 = np.random.randint(0,100,(10,10))
```

```
In [128...] arr2
```

```
Out[128...] array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
          [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
          [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
          [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
          [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
          [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
          [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
          [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
          [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
          [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

```
In [129...] arr
```

```
Out[129...] array([0, 1, 2, 3, 4, 5])
```

```
In [130...] arr[:]
```

```
Out[130...] array([0, 1, 2, 3, 4, 5])
```

```
In [131...] arr[:4]
```

```
Out[131...] array([0, 1, 2, 3])
```



```
In [132... arr2[:]
```

```
Out[132... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
        [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

```
In [133... arr2
```

```
Out[133... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
        [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

```
In [134... arr2[1,4]
```

```
Out[134... np.int32(1)
```

```
In [135... arr2
```

```
Out[135... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
        [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

```
In [136... arr2[-5,5]
```

```
Out[136... np.int32(28)
```

```
In [137... arr2
```

```
Out[137... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
        [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

In [138... arr2

Out[138... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
[41, 30, 75, 92, 1, 71, 61, 90, 14, 98],
[40, 15, 39, 23, 17, 38, 39, 83, 6, 75],
[91, 36, 1, 34, 97, 70, 7, 8, 30, 31],
[95, 76, 54, 2, 3, 15, 46, 22, 85, 88],
[87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
[29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
[64, 80, 7, 23, 61, 22, 47, 90, 36, 35],
[83, 5, 2, 24, 46, 13, 0, 47, 75, 48],
[81, 45, 30, 44, 1, 51, 3, 29, 92, 31]], dtype=int32)

In [139... arr2[-5,-5]

Out[139... np.int32(28)

In [140... arr2

Out[140... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
[41, 30, 75, 92, 1, 71, 61, 90, 14, 98],
[40, 15, 39, 23, 17, 38, 39, 83, 6, 75],
[91, 36, 1, 34, 97, 70, 7, 8, 30, 31],
[95, 76, 54, 2, 3, 15, 46, 22, 85, 88],
[87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
[29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
[64, 80, 7, 23, 61, 22, 47, 90, 36, 35],
[83, 5, 2, 24, 46, 13, 0, 47, 75, 48],
[81, 45, 30, 44, 1, 51, 3, 29, 92, 31]], dtype=int32)

In [141... arr2[-1,-2]

Out[141... np.int32(92)

In [142... arr2

Out[142... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
[41, 30, 75, 92, 1, 71, 61, 90, 14, 98],
[40, 15, 39, 23, 17, 38, 39, 83, 6, 75],
[91, 36, 1, 34, 97, 70, 7, 8, 30, 31],
[95, 76, 54, 2, 3, 15, 46, 22, 85, 88],
[87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
[29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
[64, 80, 7, 23, 61, 22, 47, 90, 36, 35],
[83, 5, 2, 24, 46, 13, 0, 47, 75, 48],
[81, 45, 30, 44, 1, 51, 3, 29, 92, 31]], dtype=int32)

In [143... arr2[::-1]

Out[143... array([[81, 45, 30, 44, 1, 51, 3, 29, 92, 31],
[83, 5, 2, 24, 46, 13, 0, 47, 75, 48],
[64, 80, 7, 23, 61, 22, 47, 90, 36, 35],
[29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
[87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
[95, 76, 54, 2, 3, 15, 46, 22, 85, 88],
[91, 36, 1, 34, 97, 70, 7, 8, 30, 31],
[40, 15, 39, 23, 17, 38, 39, 83, 6, 75],
[41, 30, 75, 92, 1, 71, 61, 90, 14, 98],
[37, 98, 39, 60, 39, 77, 17, 69, 86, 88]], dtype=int32)

In [144... arr2

```
Out[144... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
        [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

In [145... arr2[:, :-2]

```
Out[145... array([[81, 45, 30, 44,  1, 51,  3, 29, 92, 31],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98]], dtype=int32)
```

In [146... arr2

```
Out[146... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
        [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

In [147... arr2[:, :-3]

```
Out[147... array([[81, 45, 30, 44,  1, 51,  3, 29, 92, 31],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [37, 98, 39, 60, 39, 77, 17, 69, 86, 88]], dtype=int32)
```

In [148... arr2

```
Out[148... array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30],
        [64, 80,  7, 23, 61, 22, 47, 90, 36, 35],
        [83,  5,  2, 24, 46, 13,  0, 47, 75, 48],
        [81, 45, 30, 44,  1, 51,  3, 29, 92, 31]], dtype=int32)
```

In [149... arr2[:, :-3]

```
Out[149...] array([[37, 98, 39, 60, 39, 77, 17, 69, 86, 88],
        [41, 30, 75, 92,  1, 71, 61, 90, 14, 98],
        [40, 15, 39, 23, 17, 38, 39, 83,  6, 75],
        [91, 36,  1, 34, 97, 70,  7,  8, 30, 31],
        [95, 76, 54,  2,  3, 15, 46, 22, 85, 88],
        [87, 61, 22, 20, 35, 28, 49, 93, 57, 53],
        [29, 27, 11, 54, 48, 37, 47, 45, 97, 30]], dtype=int32)
```

```
In [150...] arr
```

```
Out[150...] array([0, 1, 2, 3, 4, 5])
```

```
In [151...] arr.max()
```

```
Out[151...] np.int64(5)
```

```
In [152...] arr.min()
```

```
Out[152...] np.int64(0)
```

```
In [153...] arr.max()
```

```
Out[153...] np.int64(5)
```

```
In [154...] arr
```

```
Out[154...] array([0, 1, 2, 3, 4, 5])
```

```
In [156...] from numpy import*
a = array([1,2,3,4,9])
median(a)
```

```
Out[156...] np.float64(3.0)
```

```
In [157...] # without work on import* can you please find the median ,mode
```

```
In [158...] arr
```

```
Out[158...] array([0, 1, 2, 3, 4, 5])
```

```
In [159...] arr.reshape(3,2)
```

```
Out[159...] array([[0, 1],
        [2, 3],
        [4, 5]])
```

```
In [160...] arr.reshape(6,1)
```

```
Out[160...] array([[0],
        [1],
        [2],
        [3],
        [4],
        [5]])
```

```
In [161...] arr.reshape(1,6)
```

```
Out[161...] array([[0, 1, 2, 3, 4, 5]])
```

```
In [162...] arr
```

```
Out[162...] array([0, 1, 2, 3, 4, 5])
```

```
In [163...] arr.reshape(3,2)
```

```
Out[163...] array([[0, 1],  
                  [2, 3],  
                  [4, 5]])
```

```
In [164...] arr.reshape(6,1)
```

```
Out[164...] array([[0],  
                  [1],  
                  [2],  
                  [3],  
                  [4],  
                  [5]])
```

```
In [165...] arr
```

```
Out[165...] array([0, 1, 2, 3, 4, 5])
```

```
In [168...] #indexing
```

```
mat = np.arange(0,100).reshape(10,10)
```

```
In [170...] mat
```

```
Out[170...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [174...] row = 4  
            col = 5
```

```
In [175...] col
```

```
Out[175...] 5
```

```
In [176...] row
```

```
Out[176...] 4
```

```
In [177...] mat
```

```
Out[177...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [178...] mat[row,col]
```

```
Out[178...] np.int64(45)
```

```
In [179...] mat[4,5]
```

```
Out[179...] np.int64(45)
```

```
In [180...] mat
```

```
Out[180...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [181...] mat[:]
```

```
Out[181...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [182...] col = 6
```

```
In [183...] mat
```

```
Out[183...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [184... mat[6]
```

```
Out[184... array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
In [185... mat
```

```
Out[185... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [186... # with slices
mat[:,col]
```

```
Out[186... array([ 6, 16, 26, 36, 46, 56, 66, 76, 86, 96])
```

```
In [187... mat
```

```
Out[187... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [188... mat[row,:]
```

```
Out[188... array([40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [189... mat
```

```
Out[189... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [190... mat[:,8]
```

```
Out[190... array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [191... mat[:6]
```

```
Out[191...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [192...] mat[:,8]
```

```
Out[192...] array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [193...] mat
```

```
Out[193...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [194...] mat[:,col]
```

```
Out[194...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [195...] mat[:,6]
```

```
Out[195...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [196...] row
```

```
Out[196...] 4
```

```
In [197...] mat
```

```
Out[197...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```



```
In [198... mat[:row]
```

```
Out[198... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [199... mat[row:]
```

```
Out[199... array([[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [200... mat[:]
```

```
Out[200... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [201... mat[:,8]
```

```
Out[201... array([ 8, 18, 28, 38, 48, 58, 68, 78, 88, 98])
```

```
In [202... mat[:, -1]
```

```
Out[202... array([ 9, 19, 29, 39, 49, 59, 69, 79, 89, 99])
```

```
In [203... mat
```

```
Out[203... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [204... row
```

```
Out[204... 4
```

```
In [205... col
```

```
Out[205... 6
```

```
In [206... mat[:,col]
```

```
Out[206... array([ 6, 16, 26, 36, 46, 56, 66, 76, 86, 96])
```

```
In [207... mat
```

```
Out[207... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [208... mat[1:4]
```

```
Out[208... array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [209... mat
```

```
Out[209... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [210... mat[3,-3]
```

```
Out[210... np.int64(37)
```

```
In [211... mat
```

```
Out[211... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [212... mat[6]
```

```
Out[212... array([60, 61, 62, 63, 64, 65, 66, 67, 68, 69])
```

```
In [213... mat
```

```
Out[213...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [214...] mat[5:9]
```

```
Out[214...] array([[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89]])
```

```
In [215...] mat
```

```
Out[215...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [216...] mat[0:10:3]
```

```
Out[216...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [217...] mat[0:10]
```

```
Out[217...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [218...] mat[0:10:6]
```

```
Out[218...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69]])
```

```
In [219...] mat
```

```
Out[219...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [221...] mat[:,4]
```

```
Out[221...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39]])
```

```
In [222...] mat
```

```
Out[222...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [223...] mat[:, :-2]
```

```
Out[223...] array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19]])
```

```
In [224...] mat
```

```
Out[224...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [225...] mat[:, :-5]
```

```
Out[225...] array([[90, 91, 92, 93, 94, 95, 96, 97, 98, 99],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49]])
```

```
In [226...] mat
```

```
Out[226...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [227...] mat[2:6]
```

```
Out[227...] array([[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59]])
```

```
In [228...] mat
```

```
Out[228...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [229...] mat[2:6,2:4]
```

```
Out[229...] array([[22, 23],
        [32, 33],
        [42, 43],
        [52, 53]])
```

```
In [230...] mat
```

```
Out[230...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [231...] mat[0,1]
```

```
Out[231...] np.int64(1)
```

```
In [232...] mat[0,1]
```

```
Out[232...] np.int64(1)
```

In [233...

```
mat
```

Out[233... array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9],
[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
[80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
[90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])

In [234... mat[0,1]

Out[234... np.int64(1)

In [235... mat[1,6]

Out[235... np.int64(16)

In [236... mat

Out[236... array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9],
[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
[80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
[90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])

In [237... mat[1:]

Out[237... array([[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
[80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
[90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])

In [238... mat

Out[238... array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9],
[10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
[20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
[40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
[50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
[60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
[70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
[80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
[90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])

```
In [239... mat[0:1]
```

```
Out[239... array([[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]])
```

```
In [240... mat[3:5]
```

```
Out[240... array([[30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
              [40, 41, 42, 43, 44, 45, 46, 47, 48, 49]])
```

```
In [241... mat[3,5]
```

```
Out[241... np.int64(35)
```

```
In [242... mat
```

```
Out[242... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [243... mat[1:2,2:4]
```

```
Out[243... array([[12, 13]])
```

```
In [244... mat
```

```
Out[244... array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],  
                  [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],  
                  [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],  
                  [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],  
                  [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],  
                  [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],  
                  [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],  
                  [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],  
                  [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],  
                  [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [247... mat[2:3,2:3]
```

```
Out[247... array([[22]])
```

```
In [248... mat
```

```
Out[248...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [249...] mat[2:5,2:4]
```

```
Out[249...] array([[22, 23],
          [32, 33],
          [42, 43]])
```

```
In [250...] mat
```

```
Out[250...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

masking|

```
In [256...] mat
```

```
Out[256...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
          [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
          [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
          [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
          [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
          [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
          [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
          [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
          [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
          [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [257...] id(mat)
```

```
Out[257...] 3076946938832
```

```
In [258...] mat
```



```
Out[258...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [259...] mat[mat<50]
```

```
Out[259...] array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
        34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [260...] mat[mat<=50]
```

```
Out[260...] array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
        34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50])
```

```
In [261...] mat>50
```

```
Out[261...] array([[False, False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False, False,
        False],
        [False, True,  True,  True,  True,  True,  True,  True,  True,  True,
        True],
        [ True,  True,  True,  True,  True,  True,  True,  True,  True,  True,
        True],
        [ True,  True,  True,  True,  True,  True,  True,  True,  True,  True,
        True],
        [ True,  True,  True,  True,  True,  True,  True,  True,  True,  True,
        True],
        [ True,  True,  True,  True,  True,  True,  True,  True,  True,  True,
        True]])
```

```
In [262...] mat>50
```

```
Out[262...] array([[False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True],
        [ True, True, True, True, True, True, True, True, True,
        True]])
```

```
In [263...] mat[mat==50]
```

```
Out[263...] array([50])
```

```
In [264...] mat
```

```
Out[264...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [265...] mat == 50
```

```
Out[265...] array([[False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [ True, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False],
        [False, False, False, False, False, False, False, False, False,
        False]])
```

```
In [266...] mat
```

```
Out[266...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [267...] a1 = mat[mat<50]
a1
```

```
Out[267...] array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16,
        17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33,
        34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49])
```

```
In [268...] mat
```

```
Out[268...] array([[ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9],
        [10, 11, 12, 13, 14, 15, 16, 17, 18, 19],
        [20, 21, 22, 23, 24, 25, 26, 27, 28, 29],
        [30, 31, 32, 33, 34, 35, 36, 37, 38, 39],
        [40, 41, 42, 43, 44, 45, 46, 47, 48, 49],
        [50, 51, 52, 53, 54, 55, 56, 57, 58, 59],
        [60, 61, 62, 63, 64, 65, 66, 67, 68, 69],
        [70, 71, 72, 73, 74, 75, 76, 77, 78, 79],
        [80, 81, 82, 83, 84, 85, 86, 87, 88, 89],
        [90, 91, 92, 93, 94, 95, 96, 97, 98, 99]])
```

```
In [269...] a2 = mat[mat>50]
a2
```

```
Out[269...] array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
        68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
        85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

```
In [270... a3 = mat[mat>50]
a3
```

```
Out[270... array([51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67,
        68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84,
        85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99])
```

```
In [271... a4 = mat[mat==50]
a4
```

```
Out[271... array([50])
```

```
In [283... def wish():
    print('good even')

wish()

def wish():
    print('good even')

wish()

def wish():
    print('good even')

wish()

def wish():
    print('good even')

wish()
wish()
wish()

list1 = ['a', 'b', 'g', 1, 5]

print(list1.pop())    # removes last item

x = [1, 2, 3]

y = x.copy()

x.append(4)

print(x)
print(y)
```

```
good even  
good even  
good even  
good even  
good even  
good even  
5  
[1, 2, 3, 4]  
[1, 2, 3]
```

In []: